

JAVIER CHUA BING YAN

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Aug 2024 – May 2027

Bachelor of Computing (Computer Science)

- Relevant Coursework: System Design, Operating Systems, Networks, Database Systems, Machine Learning, Distributed Systems, Computer Security, Software Engineering

TECHNICAL SKILLS

Languages: Java; JavaScript/TypeScript; Python; Bash/Shell; C; C++; Dart

Web & Mobile: Next.js; React; Node.js; Express; React Native; Flutter; Expo; Vite; NextAuth; Streamlit; REST APIs

Cloud & DevOps: AWS (ECS, ECR, ALB, SSM); Terraform; Docker; Nginx; Google Cloud Run; Argo CD; Azure DevOps; CI/CD; Kafka

Data & Databases: PostgreSQL; MySQL; MongoDB; SQLite; H2; Prisma; Pandas; NumPy; scikit-learn; InfluxDB3

Security, AI & Tools: SonarQube; Trivy; Entra ID (OIDC); LangGraph; LangChain; CatBoost; SevOne; Vitest; Jest; JUnit; Pytest

EXPERIENCE

VISA

Network & Security Engineering Intern

May 2026 – Present

- Tech Stack:** Python, LangGraph, CatBoost, scikit-learn, SevOne, NetBrain, Netmiko, MongoDB, Kafka, InfluxDB3, ServiceNow
- Designing requirements and data specifications for a hybrid ML + LLM network outage prediction agent; Defining the offline training pipeline from SevOne telemetry through Parquet normalisation, feature engineering (rolling window statistics, rate-of-change, time-since-threshold-crossing), Isolation Forest unsupervised labelling, and CatBoost supervised classification
- Specifying a LangGraph StateGraph agent covering telemetry fetch, CatBoost anomaly scoring, NetBrain topology enrichment, GPT-4o event correlation, confidence-gated human-in-the-loop escalation, and automated ServiceNow incident creation
- Developing Python self-heal automation scripts using Netmiko for Arista EOS and Cisco IOS devices; Integrating NetBrain REST API for real-time topology queries and event-driven remediation task execution

CENTRAL PROVIDENT FUND BOARD (CPFBoard)

Software Engineering Intern

Jan 2026 – May 2026

- Tech Stack:** TypeScript, Node.js, Next.js, AWS, Terraform, Docker, Nginx, Azure DevOps, SonarQube, Trivy, Vitest
- Implemented a zero-rebuild ECS rollback mechanism by querying ECR manifests and forcing immediate cluster state reconciliations, enabling sub-minute incident recovery
- Built a dynamic Infrastructure-as-Code provisioning engine with Terraform, isolating zero-trust multi-tenant projects within a shared AWS ECS and ALB architecture
- Developed an automated multi-step project orchestration pipeline, reducing manual setup time from days to seconds
- Engineered a shift-left DevSecOps pipeline featuring automated SonarQube quality gating and dynamic Trivy container scanning with automatic hardened base-image injection

PROJECTS

city.ai

Jan 2026 – Mar 2026

- Tech Stack:** Python, React 19, Cloudflare Pages, Supabase (pgvector), Gemini 2.5 Flash, Cloudflare Tunnel, Zod, Vitest
- Architected serverless AI orchestration on Cloudflare Pages Functions for multi-turn Gemini 2.5 Flash conversations; Enforced V8 isolate runtime parity via Cloudflare Tunnel
- Engineered a Python RAG ingestion pipeline indexing 1,700+ hyperlocal data chunks into Supabase pgvector with hybrid rate-limiting; achieved 100% embedding success

Java Blockchain

Jul 2025 – Sep 2025

- Tech Stack:** Java, SHA-256, Concurrency, H2, HikariCP, Gson, Maven
- Architected a concurrent Proof-of-Work mining engine with a custom ExecutorService thread pool with atomic primitives, distributing nonce exploration across cores to accelerate block generation

Splend – Group Travel Management iOS App

May 2025 – Aug 2025

- Tech Stack:** React Native, TypeScript, Firebase Auth/Firestore, Google Cloud Run (OpenAI GPT-4o Vision), Jest
- Automated receipt extraction using camera/gallery upload with GPT-4o Vision on Google Cloud Run; enforced transaction integrity through atomic Firestore batch writes

LEADERSHIP EXPERIENCE

KENT RIDGE HALL

Touch Rugby (M) Captain

Jul 2025 – May 2026

- Led Inter-Hall Games squad strategy and spearheaded technical training regimes

Hall Promotion Board (HPB), Head of Finance

Jul 2024 – Jul 2025

- Led cross-functional team of 8 members in budgeting and procurement for Open House Day and HONUS Day 2025

SINGAPORE ARMED FORCES (9TH DIVISION)

Infantry Officer, 1st Lieutenant

Mar 2023 – Jan 2024

- Commanded and trained a 40-man platoon; coordinated tactical plans and delivered mission-critical briefings to senior leadership